

Project Design Phase-II

Technology Stack (Architecture & Stack)

Date 31 January 3035

Team ID SWTID-2026-4968

Project Name Script - controlled ACL - Restrict Record Access Based On Field Value

Maximum Marks 4 Marks

Technical Architecture:

The Deliverable shall include the architectural diagram as below and the information as per Table 1 & Table 2.

Example: Script-controlled Access Control List (ACL) that restricts record access based on a field value.

Reference: ServiceNow Access Control (ACL) and server-side scripting documentation.

Table-1 : Components & Technologies:

S.No	Component	Description	Technology
1	User Interface	How user interacts with the application, such as forms, lists and record views.	ServiceNow UI / Now Experience
2	Application Logic-1	Script logic evaluates the target record field value before granting or denying access.	JavaScript / ServiceNow Server-side Script
3	Application Logic-2	ACL condition and script determine whether the current user can access the record.	ServiceNow ACL Script / GlideRecord
4	Application Logic-3	Role and user-context checks can be combined with field-value validation.	ServiceNow User / Role APIs
5	Database	Stores records and the field values used by the access-control logic.	ServiceNow Platform Database
6	Cloud Database	Database service supporting the ServiceNow application platform.	ServiceNow Cloud Platform
7	File Storage	Stores attachments and related application files where applicable.	ServiceNow Attachment Storage
8	External API-1	Optional integration for obtaining user, identity or business information.	REST API / ServiceNow REST Integration
9	External API-2	Optional external service used when field-based access depends on external data.	REST / JSON API
10	Machine Learning Model	Not required for the core field-value ACL implementation.	Not Applicable
11	Infrastructure (Server / Cloud)	Application deployment and execution on the ServiceNow platform.	ServiceNow Cloud / SaaS Platform

Table-1 : Components & Technologies (continued)

S.No	Component	Description	Technology
2	Application Logic-1	Field-value validation and access decision logic.	JavaScript / ACL Script
3	Application Logic-2	Record-level authorization based on the requested operation.	ServiceNow ACL
4	Application Logic-3	User and role checks used with the field-value rule.	GlideSystem / User Roles

Table-2: Application Characteristics:

S.No	Characteristics	Description	Technology
1	Open-Source Frameworks	List the open-source frameworks used. The core ACL implementation is platform-specific rather than dependent on an external open-source framework.	JavaScript; ServiceNow scripting APIs
2	Security	Restrict record access by evaluating a field value together with the current user's access context. ACL rules prevent unauthorized read/write access.	ServiceNow ACL; Roles; Conditions; Server-side Script
3	Scalable Architecture	Authorization logic is executed within the platform's access-control layer and can be applied consistently across records and operations.	ServiceNow Platform; ACL Framework
4	Availability	The application runs on the ServiceNow cloud platform, with access-control evaluation handled as part of normal platform request processing.	ServiceNow Cloud / SaaS

Table-2: Application Characteristics (continued)

S.No	Characteristics	Description	Technology
5	Performance	The ACL script should use simple field-value checks and avoid unnecessary database queries so that authorization evaluation remains efficient.	JavaScript ACL Script; GlideRecord only when required

References:

1. ServiceNow documentation – Access Control Lists (ACLs).
2. ServiceNow documentation – Access Control rule scripting and conditions.
3. ServiceNow documentation – GlideRecord and server-side scripting APIs.
4. ServiceNow documentation – Users, groups and roles.